

Lahiru Wijesundara

22048517

UFCFLK-60-M Creative Technology Dissertation

MSc Commercial Games Development

Assessment 1 – Dissertation Proposal

A Comparative Study of DQN and PPO Algorithms in 2D Fighting Game

Using Unity ML-Agents

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## 1. INTRODUCTION

Artificial Intelligence (AI) applications are expanding as their needs increase in the modern world (Jiang, 2020). AI is used in many areas, such as object recognition, medical research, translation, speech recognition, personalisation, etc. Deep reinforcement learning (DRL) in video games is also experiencing rapid growth in AI. Video games introduce various kinds of challenges in their nature of complexity, which includes interactions between agents and their environment, as well as those involving players and agents, which are often unpredictable. AI has proven to be more skilled than most human players in games that follow a turn-based format, such as chess and card games (Liang, Li, 2022). These games are considered games with a fixed environment; however, the same level of performance has yet to be achieved in fighting games. Fighting games present significant challenges due to their limited decision-making time, extensive decision space, and diverse strategies. Researchers should customise each fighting game genre's design and development to overcome these challenges.

Despite the advancement of DRL in video games, more information is needed on the comparative analysis of different DRL algorithms in the 2D fighting game genre. Although Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) have demonstrated potential in other applications, such as those documented by (Yoon, Kim, 2017) and (Liang et al., 2021), there remains a need for an extensive comparison of their performance within this area. This presents an excellent research opportunity to significantly widen the understanding of how these algorithms perform in highly dynamic environments regarding learning efficiency and win rate.

This research proposal proposes the design, development, and comparative analysis of AI for the 2D fighting game using DQN and PPO algorithms. The research methodology proposed by Ramlan et al. (2021) will be used as a baseline to design a 2D game and to model AI agents. All the gameplay mechanics will be implemented using the Unity game engine by utilising its capabilities, such as the physics system, high-level functionalities and rendering system. Unity's open-source ML-Agent toolkit provides seamless integration to develop and test advanced RL algorithms such as DQN and PPO. A new reward-shaping-based self-play method proposed by Oh et al. 2022 will be used to train AI agents.

Main Research question:

Despite the advances, a comparative analysis of different DRL algorithms, particularly DQN and PPO, in the 2D fighting games genre needs to be explored.

Sub-research questions:

- How to develop a one-on-one 2D fighting AI agent using the Unity game engine
- How to train the AI agents with DQN and PPO algorithms separately using the Unity ML-Agents toolkit.

- How to compare the performance of these algorithms in terms of learning efficiency and win rate.

### **Aim:**

This research centres on the comparative performance analysis of DQN and PPO algorithms in a challenging 2D fighting game environment. The research aims to address the gap present in the current literature review. Such a comparison is essential in the 2D fighting game genre to identify how the different DRL algorithms perform regarding learning efficiency and win rate in an overly complex, real-time decision-making environment. This project's primary artefact will be a 2D fighting game with trained AI agents using DQN and PPO, which will be used as a testbed to make comparative performance analyses regarding DRL algorithms. Even though this research contributes to AI in gaming, it offers insight into how to design and develop a complete 2D fighting game in the Unity game engine.

### **Objectives:**

To achieve the research aims, the project starts with designing and implementing a 2D dynamic fighting game environment using the Unity game engine, which serves as AI agents' training ground. The next step is to implement and train DQN and PPO algorithms separately by utilising the Unity ML-Agents toolkit. The agent training process will be done through hyperparameter tuning and the reward-shaping-based self-play method. After training, performance metrics such as learning efficiency and win rate will be established to compare each algorithm.

## **2. BACKGROUND**

### **2.1. Deep Reinforcement Learning**

Reinforcement learning is an AI method that teaches agents to improve their decision-making with environmental feedback, and the goal is to maximise the agents' rewards over time (Krichen, 2023). According to the author, Reinforcement learning (RL) methods traditionally need to be revised when dealing with high-dimensional state and action spaces as they depend on manually designed features and linear function approximation. However, introducing Deep Reinforcement Learning (DRL), which uses Deep Neural Networks (DNNs) to represent an agent's policy and value function, leads to state-of-the-art performance in many applications. Jiang (2020) discussed DRL in games and examined achievements and challenges in various game types. He analyses challenges such as sample efficiency, exploration, and exploitation trade-offs, and delayed and sparse rewards. According to the author, DRL algorithms led to significant accomplishments in games such as AlphaGo, AlphaGo Zero, MuZero in Atari games, and an unnamed AI in the first-person shooter game Doom. In addition, DRL algorithms emerged victorious against professional players in the real-time strategy game Starcraft II (Vinyals O. et al., 2019).

According to Ramlan et al. (2021), a finite state machine is a common technique for developing AI in video games utilising predefined actions. However, a significant drawback of this approach is that those actions are easily predictable. So, it can lead to tedious gameplay, making players lose interest in the game as they already know how the AI will behave. However, an AI trained with a DRL can be more challenging and exciting for the player.

## **2.2. AI in Fighting Games**

The research area of AI in games focuses on how AI agents interact with their environment and how to play like human players or even better (Jiang, 2020). Different game genres offer diverse types of challenges. The fighting game genre is recognised as one of the most complex types of player-versus-player (PVP) games (Oh et al., 2022). In 2013, a new AI competition was explicitly introduced for 2D fighting games, FightingICE (FICE) (Lu et al., 2013), and Monte Carlo tree search-based methods and other algorithms have been applied in this competition over the years (Ishihara et al., 2018; Yoshida et al., 2016). Ritsumeikan University developed FightingICE, and its unique features, such as the 1v1 fighting game environment, make it an excellent platform for studying AI algorithms. Unlike board game AIs such as Deep Blue or AlphaGO, fighting games have limited time for AI reactions, making them more challenging. On the other hand, they have shorter playtimes and less action complexity than real-time simulation games, including Starcraft or DOTA2, making them less computationally demanding (Kim, Park, Yang, 2020). Cruz and Uresti (2018) developed a framework using reinforcement learning to produce non-playable characters (NPCs) that are more convincing in video games. Using Street Fighter IV, a 2D fighting game, they tested this framework and achieved promising results. According to Oh et al. (2022), although DRL succeeded in many other game genres, no further study could outperform human players in a complex fighting game until their research. Using an RL method and innovative self-play techniques, they developed 1v1 battle AI agents for the 3D Blade and Soul fighting game.

## **2.3. Deep Q-Networks and Proximal Policy Optimization in AI Agents**

A novel AI algorithm called Deep Q-Network was introduced by Mnih et al. (2015), which employs end-to-end reinforcement learning to identify effective policies directly from high-dimensional sensory inputs. DQN integrates Q-learning with deep neural networks to guide agents' decisions by approximating the Q-value. Q value represents the expected rewards. The algorithm achieved a level of control in playing 49 Atari 2600 games on par with human performance, using only the pixels and game score as inputs. This performance surpassed that of all previous algorithms. According to Takano et al. (2019), DQN works well with fighting games but is only sometimes superior to a new opponent, which was unseen before. The author suggests a self-play-based training method to overcome this drawback while keeping the algorithm's strong points.

Proximal Policy Optimization was introduced by OpenAI in 2017 to retain the advantages of Trust Region Policy Optimization (TRPO) while improving sample efficiency and simplifying implementation (Schulman et al., 2017). PPO is categorised as one of the policy gradient methods

in RL (Melo, Maximo, 2019). Policy gradient methods aim to create a parameterised policy that can directly make decisions from the state space. During training, they estimate the policy gradient and use a stochastic gradient ascent algorithm. PPO is a widely used algorithm in the fighting game genre. In 2021, the FightingICE competition winners under the speed running league and the self-play category used the PPO algorithm to develop their agents (Nimpattavong, Choensawat, Sookhanaphibarn, 2022). Liang et al. (2021) introduced a new method called ERHEAPPO, which combines PPO and the horizon evolution algorithm (ERHEA). They compared it with the 2020 fighting game artificial intelligence competition (FTGAIC) champion, and the results were promising. Also, Ramlan et al. (2021) trained a 2D fighting AI agent with a PPO algorithm. They found that reducing the learning rate during DRL could enhance the agent's cumulative reward.

## **2.4. Self-play reinforcement learning**

Self-play reinforcement learning is a process where agents learn by competing against their own copies (Liu et al., 2021). The winning agent is replaced with a copy of itself in the training process, which creates a perfect training environment and gives the agent strong competitors to play against. Self-play reinforcement learning benefits game agents such as AlphaGo Zero (Silver et al., 2017) and AlphaZero (Silver et al., 2016). These agents have demonstrated the capability to surpass human players in perfect-information games, attributed to the self-play reinforcement learning technique. Kim, Park, and Yang (2020) presented a method to develop AI agents using self-play and Monte Carlo Tree Search (MCTS) and tested MCTS and self-play in a 1:3 ratio with top-ranking AI agents of FightingICE competition in 2017 and 2019. They were able to achieve a 94.4%-win rate. Oh et al. (2022) developed three sets of agent groups, aggressive, balanced, and defensive, using reward shaping. Agents were regularly saved in a shared pool and utilised for self-play learning. The selection of agents from the shared pool was based on probabilities, with the more recent models having a higher likelihood of being selected. According to the above authors, the trained AI agents played against five professional players of the Blade and Soul commercial game and achieved a 62% winning rate.

## **2.5. Unity ML-Agents Toolkit**

An interface is provided by the Unity Machine Learning Agents Toolkit (ML-Agents) to train RL agents in interactive content made with Unity 3D (Farrapo et al., 2022). Researchers can use a Python API to communicate with the engine, which enables them to implement agents using frameworks such as TensorFlow and PyTorch. The toolkit is equipped with pre-built DRL techniques and scenarios, quickly making testing environments and agents easy. PPO and Soft Critic (SAC (Soft Actor Critic)) natively support DRL algorithms, and other algorithms like imitation learning (IL) and intrinsic curiosity module (ICM) are available to encourage inquisitive exploration (Rauch, Korečko, Gazda, 2022). All these algorithms are executed using a locally running Python backend server. Also, it supports integrating custom algorithms, which helps expand the possibilities of using the environment beyond what is natively offered by Unity.

According to Ramlan et al. (2021) used this toolkit to train a 2D fighting game AI agent using the PPO algorithm.

Zakharenkov and Makarov (2021) conducted a DQN vs PPO comparison in a first-person shooter genre, and the results show that the DQN model performs better than the PPO model in handling the challenging responsibility of navigation and decision-making. Based on the literature, in the 2D fighting games genre, DRL algorithms, notably DQN and PPO algorithms, have been used to train AI agents. However, proper comparative analysis of DQN and PPO algorithms in the 2D fighting game genre has been lacking until now, and this research proposes a methodology to fill that gap.

### **3. METHODOLOGY**

This section proposes a method to design, develop, and train a 2D fighting game AI agent using the DQN and PPO algorithms and how to compare their performance.

#### **3.1. Design and develop 2D fighting game.**

Designing and developing a 2D fighting game environment involves careful consideration of gameplay mechanics and AI integration capabilities, which will then be used as the basis for training and evaluating the AI agents. The game development methodology suggested by Ramlan et al. (2021) will be used to develop the game environment, and based on that, here is a detailed breakdown of the process involving the Unity game engine.

Game Environment Features:

- Characters

The game will consist of two AI agents with health and stamina, and the available game states for each agent are mentioned below.

- Idle state
- Move State (Move left or right)
- Slash attack state

During a fight, both agents keep a check on their own health and stamina levels, as well as those of their opponent, while also being mindful of the distance between them. At the start of every match, an equal amount of health and stamina is given to both parties.

- Arena

The 2D arena has boundaries on the outer edges to ensure that the fighters do not move out of the playing area. Clear visual indicators(bars) show the current health and stamina in the user interface (UI).

- Game Mechanics and Physics

It is required to create a fundamental physics system that can handle movements and interactions. This system should be capable of detecting collisions during attacks and implement a basic health system, which can reduce the health points when a successful hit is detected.

### 3.2. AI model development for the agents

The model for the AI agents will be developed based on the DRL network architecture proposed by Ramlan et al. (2021) using Unity's ML-Agents toolkit. During the learning stage, the agent carefully observes its surroundings. This includes monitoring its health, stamina, and the distance between itself and its opponent. Based on these observations, the agent decides whether to move or attack. If the action taken by the agent results in a favourable outcome, such as hitting its opponent, then a positive reward is given. However, if the action taken by the agent results in an unfavourable outcome, then a negative reward is presented instead.

There will be six input neurons in DNN (deep neural networks), including the agent's health, stamina, opponent's health, stamina, opponent's direction, and the distance between the agent and the opponent. The DNN will also consist of three hidden layers containing ten neurons. The output layer will have two neurons, one for movement and the other for attack. The reward function is defined based on environment values (player's and opponent's health and stamina points, distance between them), and the aim is to provide feedback to the agent so the agent can learn. The proposed reward allocation by Ramlan et al. (2021) is shown below.

| Action                     | Reward |
|----------------------------|--------|
| Move close to the opponent | +0.1   |
| Hitting the opponent       | +5     |
| Perform attack but missed  | -5     |
| Win a fight                | +50    |
| Lose the fight             | -50    |

### 3.3. Model training and comparative performance analysis.

The methodology to train the agents is based on the innovative self-play approach presented by Oh et al. (2022). The agents' learning will be executed across three distinct fighting styles: aggressive, defensive, and balanced. Subsequently, the agents will engage in battles against each other. Each fighting style is determined by reward sharpening, based on three dimensions of rewards, as mentioned below.

- Time penalty - Penalties are imposed more severely on the hostile agent for every turn, motivating it to finish the game more rapidly.



- Health Point (HP) ratio - Aggressive players prioritise reducing the opponent's HP over preserving their own HP, whereas defensive players typically prioritise maintaining their own HP over reducing the opponent's.
- Distance penalty - Players who adopt a defensive strategy tend to maintain a specific distance from their opponents to be able to react appropriately when attacked. In contrast, more aggressive players tend to close in on their opponents and launch relentless attacks.

The following table shows reward details for each fighting style proposed by Oh et al. (2022).

|                  | Aggressive | Balanced | Defensive |
|------------------|------------|----------|-----------|
| Time penalty     | 0.008      | 0.004    | 0.0       |
| HP ratio         | 5:5        | 5:5      | 6:4       |
| Distance penalty | 0.002      | 0.002    | 0.0       |

The method Ramlan et al. (2021) suggested for analysing the DRL algorithm will be utilised to evaluate model performance. The evaluation will be based on the cumulative reward attained during each iteration. Modifying parameters is crucial for training the model to achieve the desired performance and expedite the learning model's convergence. The parameters that are supposed to be adjusted in this project are the number of neurons in the hidden layers and the learning rate. Hidden layers correspond to the number of neurons present in each hidden layer. The learning rate determines how the model learns about the given problem. To determine the win rate between the DQN agent and PPO agent, AI agents play against each other a thousand times. This method was previously employed by Oh et al. (2022) to display the impact of the suggested self-play curriculum in their study.

### 3.4. Deliverables

All of the research project's deliverables are mentioned below.

- Demo
  - A demo version of the 2D fighting game with AI bots trained using DQN and PPO
- Project repository
  - GitLab repository containing all the code, documentation, models, and other relevant materials generated during the project. This facilitates sharing, collaboration, and future research extensions.
- Presentation Materials
  - Presentation slides for the final viva presentation.

### 3.5. Hardware and Software Requirements

Since the proposed project is based on 2D game development and training DRL algorithms within the Unity game engine and ML-Agents Toolkit, it requires hardware and software resources capable of supporting game development, machine learning development, training, and simulation.

- Hardware
  - Central Processing Unit (CPU):
    - High-performance CPU with multiple cores (Intel Core i7 or higher) to manage the computational load of training models.
  - Graphics Processing Unit (GPU):
    - A dedicated GPU is crucial for deep learning tasks. NVIDIA GPUs (Graphics Processing Unit) with CUDA support (NVIDIA RTX 3070 or higher) to accelerate model training times.
  - Memory (RAM):
    - At least 16GB of RAM is preferable for more extensive model training.
  - Storage:
    - Solid State Drive (SSD) with at least 512GB of storage for faster read/write speeds, which is beneficial for handling machine learning models. Additional HDD storage may be helpful for backups.
  - Network:
    - A stable and fast internet connection is needed to access online resources and collaborate with version control systems.
- Software
  - Operating System:
    - Windows 11
  - Development Environment:
    - Unity Engine: For creating the 2D fighting game environment. The latest stable version (2022.3 or later) is compatible with the ML-Agents Toolkit.
    - Unity ML-Agents Toolkit: For integrating machine learning models into the Unity environment.
  - Machine Learning and Deep Learning Libraries:
    - TensorFlow and/or PyTorch: Primary deep learning libraries used for implementing DQN and PPO algorithms.
  - Integrated Development Environment (IDE):
    - Visual Studio
  - Version Control:
    - GitLab

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## 5. APPENDICES

| Task                                                         | 26-Feb |  |  | March |       |       | April |       |       | May   |       |       | June  |       |       | July  |       |       | 29-Aug |       |       |       |       |       |       |       |       |       |       |       |
|--------------------------------------------------------------|--------|--|--|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                              |        |  |  | WK 01 | WK 02 | WK 03 | WK 04 | WK 05 | WK 06 | WK 07 | WK 08 | WK 09 | WK 10 | WK 11 | WK 12 | WK 13 | WK 14 | WK 15 | WK 16  | WK 17 | WK 18 | WK 19 | WK 20 | WK 21 | WK 22 | WK 23 | WK 24 | WK 25 | WK 26 | WK 27 |
| Supervisor meetings                                          |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Study deep reinforcement learning & Unity ML-Agents toolkit  |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Documentation & 2D Fighting Game Development                 |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Prepare and update GDD and TDD                               |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| 2D asset integration and level design/development            |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Game mechanics and UI implementation for fighter agent       |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Testing and Troubleshooting                                  |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| AI Agents Development Using Unity ML Agent Toolkit           |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| PPO agent implementation and training                        |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| DQN agent implementation and training                        |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Testing and Troubleshooting                                  |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Comparative analysis                                         |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Finalisation                                                 |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |
| Finalising the project, viva slides preparation & submission |        |  |  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |       |       |       |       |       |       |       |       |       |       |       |

Figure 1: Dissertation plan from 26 February to 29 August.

## 6. ETHICAL APPROVAL FORM



Faculty of Environment & Technology  
Faculty Research Ethics Committee (FREC)

### Ethical Review Checklist for Undergraduate and Postgraduate Modules

*Please provide project details and complete the checklist below.*

#### Project Details:

|                        |                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------|
| Module name            | Creative Technology Dissertation                                                        |
| Module code            | UFCFLK-60-M                                                                             |
| <b>Module leader</b>   | Dr Nikolaos Ersotelos                                                                   |
| Project Supervisor     | Mrs Beth Mackey                                                                         |
| Proposed project title | A Comparative Study of DQN and PPO Algorithms in 2D Fighting Game Using Unity ML-Agents |

#### Applicant Details:

|                         |                                         |
|-------------------------|-----------------------------------------|
| Name of Student         | Lahiru Wijesundara                      |
| Student Number          | 22048517                                |
| Student's email address | Wijesundara2.Wijesundara@live.uwe.ac.uk |

| CHECKLIST QUESTIONS |                                                                                                                                                     | Yes/No | Explanation                                                                                    |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------|
| 1.                  | Does the proposed project involve human tissue, human participants, animals, environmental damage, the NHS, or data collected outside the UK?       | No     | <i>If the answer to this is 'No' then no further checks in the list need to be considered.</i> |
| 2.                  | Will participants be clearly asked to give consent to take part in the research and informed about how data collected in the research will be used? |        |                                                                                                |
| 3.                  | If they choose, can a participant withdraw at any time (prior to a point of "no return" in the use of their data)? Are they told this?              |        |                                                                                                |

| CHECKLIST QUESTIONS |                                                                                                                                                         | Yes/No | Explanation |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|
| 4.                  | Are measures in place to provide confidentiality for participants and ensure secure management and disposal of data collected from them?                |        |             |
| 5.                  | Does the study involve people who are particularly vulnerable or unable to give informed consent (eg, children or people with learning difficulties)?   |        |             |
| 6.                  | Could your research cause stress, physical or psychological harm to humans or animals, or environmental damage?                                         |        |             |
| 7.                  | Could any aspects of the research lead to unethical behaviour by participants or researchers (eg, invasion of privacy, deceit, coercion, fraud, abuse)? |        |             |
| 8.                  | Does the research involve the NHS or collection or storage of human tissue (includes anything containing human cells, such as saliva and urine)?        |        |             |
| 9.                  | Will the research require any UWE staff or students to collect primary data from outside the UK?                                                        |        |             |

Your explanations should indicate briefly for Qs 2-4 how these requirements will be met, and for Qs 5-9 what the pertinent concerns are.

- Minimal Risk: If Q 1 is answered 'No', then no ethics approval is needed.
- Low Risk: If Qs 2-4 are answered 'Yes' and Qs 5-9 are answered 'No', then no approval is needed from the *Faculty Research Ethics Committee* (FREC). However, your supervisor must approve (a) your information and consent forms (Qs 2 & 3) and (b) your measures for participant confidentiality and secure data management (Q4).
- High Risk: If any of Qs 5-9 are answered 'Yes', then you must submit an application for full ethics approval *before* the project can start. This can take up to 6 weeks. Consult your supervisor about how to apply for full ethics approval.

Your supervisor must check your responses above *before* you submit this form.

Submit this completed form via the *Assignments* area in Blackboard (or elsewhere if so directed by the module leader or your supervisor).

After you have uploaded this form, your supervisor will confirm it has been correctly completed by “marking” it as *Passed*/100% via the *My Grades* link on the Blackboard.

Further research ethics guidance is available at <http://www1.uwe.ac.uk/research/researchethics>